Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Production LLM & Agentic Systems · LLM Evaluation & Monitoring · Python 9yr · Anthropic / OpenAI / Groq · M.S. Data Analytics

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PROFESSIONAL SUMMARY

AI engineer with 1+ year of production LLM and agentic systems work, built on 9 years of full-stack production engineering. I built AgentProbe -- an LLM evaluation and monitoring framework with a 5-metric CI-integrated eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), a regression gate blocking deploys on quality degradation, and LLM-as-judge scoring across 1,240+ test cases. That directly answers Schwab AI.x's requirement for "evaluating and monitoring LLM-generated outputs for quality and consistency." I also built LeaseI: a production 8-agent LangGraph agentic platform (0→1 lifecycle) -- the AI system that AgentProbe evaluates. I work across Anthropic/Claude, OpenAI, and Groq APIs. Python is my primary language (9yr). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Portfolio: venkatalakshmiparimala.github.io/Portfolio/about.html

TECHNICAL SKILLS

AI, LLM & Evaluation: LLM APIs (Anthropic/Claude, OpenAI, Groq, Ollama), LangGraph (multi-agent orchestration), Agentic AI, Production AI Systems, LLM Evaluation, LLM Monitoring, LLM-as-Judge, Hallucination Detection, Evaluation Framework Design, Observability, Agent Tracing, Confidence Scoring, HITL, Guardrails, RAG Pipelines, Tool Orchestration, Prompt Engineering, Context Management, Multi-Agent Systems, LangChain, Embeddings, Document Intelligence, OCR, Claude Code (daily)

Languages & Frameworks: Python (9yr production), TypeScript, JavaScript, SQL/PL-SQL, R, Bash/shell scripting, HTML/CSS; FastAPI, Flask, Angular, React.js

Cloud, Data & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, Kubernetes, Containers, CI/CD (GitHub Actions), Infrastructure as Code, Cloud-Native Applications, GenAI Application Delivery, PostgreSQL (200+ queries optimized, stored procedures, multi-tenant schemas), MySQL, Oracle, ETL/ELT Pipelines, Data Modeling, Large Dataset Pipelines, REST APIs, Microservices, RabbitMQ, Automated Testing, Regression Testing, Git, Agile/Scrum; Machine Learning: PyTorch, scikit-learn, XGBoost, Random Forest, Pandas, NumPy

EXPERIENCE

Entrata -- Multifamily Real Estate SaaS

Sep 2018 -- Aug 2024

Senior Software Engineer -- Full Stack & Data Pipelines · Python · PostgreSQL · AWS · Lehi, UT (Remote)

- **Large dataset data pipelines with production reliability requirements** -- sole owner of annual HUD income-limit import pipeline: few thousand federal source rows expanded to ~180,000 domain records powering eligibility, rent calculations, and recertification for 100K+ housing units. Pre-import validation, reconciliation, and rollback on every release -- a wrong import propagates to thousands of compliance decisions. Zero data integrity violations across full 7-year ownership period.
- **Cloud-native production systems at scale** -- AWS S3 migration with Libsodium-encrypted ETL pipelines (100% PHI compliance, zero security incidents); RabbitMQ async processing with idempotency guarantees; optimized 200+ SQL queries (30% performance gain) through execution plan analysis; Docker microservices redesign (50% scalability improvement, 30% deployment reduction); CI/CD across 120+ Agile sprints.
- **Full-stack delivery and engineering culture** -- shipped 20+ full-stack production modules (Python, JavaScript, PostgreSQL); 300+ peer code reviews; mentored 5+ engineers; built debugging tools reducing time-to-resolution 40%; 99.5% uptime on compliance-critical systems.

George Mason University

Aug 2024 -- May 2026

Research Associate & Graduate Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); applied Isolation Forest to automate data integrity checks across 3 research pipelines; designed Tableau/ggplot2 dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and data engineering.

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer -- Agentic Systems & LLM Application Engineering

- Built an LLM-powered agentic application (LangGraph stateful graph) over a regulatory corpus with RAG retrieval, validation guardrails, and conditional routing -- every compliance decision grounded in cited regulation. Multi-step reasoning loop with graceful failure handling; LLM output validated against deterministic rules before export.

Entrata India -- Pune

Jan 2017 -- Sep 2018

Associate Software Engineer -- Python · PostgreSQL · Full Stack

- Built property management modules; income-limit import pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration; 56 state-specific compliance document templates.

PROJECTS

AgentProbe: LLM Evaluation & Monitoring Framework -- CI Quality Gate for Production AI

LLM Evaluation · Monitoring · Hallucination Detection · CI/CD · Multi-Provider · React

Python · FastAPI · React.js · TypeScript · ChromaDB · GitHub Actions · Anthropic/Claude · OpenAI · Groq | github.com/venkatalakshmiparimala/AgentProbe

Production LLM evaluation and monitoring framework -- what Schwab AI.x's "evaluating and monitoring LLM-generated outputs for quality and consistency" preferred qualification describes. **5-metric automated evaluation engine:** correctness, groundedness, latency, token efficiency, hallucination rate -- each metric tracked across 1,240+ test cases with multi-trial sampling to reduce variance. **CI regression gate:** GitHub Actions integration blocks deploys when quality scores degrade -- I don't ship a prompt change without knowing what it does across 1,240 conversations. **LLM-as-judge:** structured scoring with configurable rubrics; results tracked across eval runs for trend analysis. **Multi-provider routing:** Anthropic/Claude, OpenAI, Groq -- used to evaluate cost vs. quality tradeoffs across providers. 15-endpoint FastAPI backend + React/TypeScript frontend.

Leasel: Production Agentic AI System -- 0→1 LLM Application Lifecycle

LangGraph · Production AI · RAG · Agentic Orchestration · HITL · FastAPI

Python · LangGraph (8-agent StateGraph) · FastAPI · TF-IDF RAG · SQLite · OCR · GitHub Actions | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production LLM-powered agentic application built 0→1 -- the full lifecycle: prompt design, architecture, deployment, monitoring via AgentProbe, iteration. **Architecture:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Reasoning → Validation → HITL Escalation) -- planning loop, multi-step tool calls, conditional routing, graceful degradation (confidence drop → human review). **Retrieval:** TF-IDF RAG (deliberate over vector DB -- reliable for compliance retrieval; "pragmatic about models"). **Quality gate:** AgentProbe evaluates this system; CI regression gate prevents quality regressions from shipping. **Results:** 93.6% precision, 87.9% F1 across 1,240 test cases; pass@1 81.7%, pass@3 88.3%.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · React · LLMs · BERT · Full Stack · HIPAA

Python · Flask · React · Docker · spaCy · BERT · Embeddings · FHIR API | github.com/venkatalakshmiparimala/AI_on_FHIR

HIPAA-compliant Flask + React full-stack platform -- NLP + LLM + BERT translating natural-language queries into regulated FHIR API calls; reduced manual clinical retrieval time ~60%; enforced RBAC across structured medical records. Full-stack AI application delivery: Python backend + React frontend + Docker.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2013--2017